

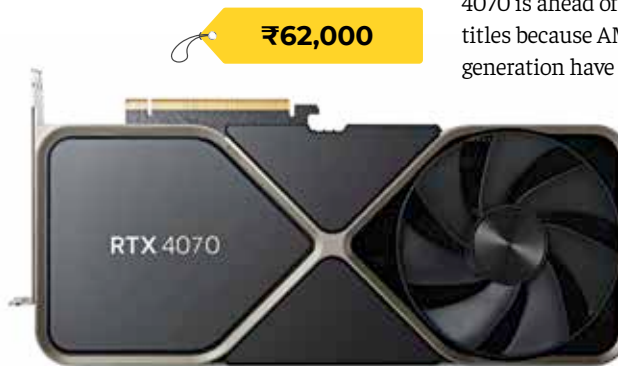
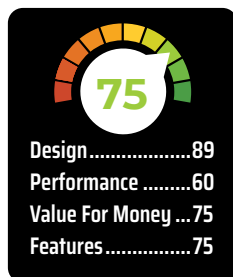
NVIDIA GeForce RTX 4070

A slightly improved RTX 3080

The RTX 4070, given its pricing (INR 62,000), is geared towards the higher end of the mid-range segment, which is where you find cards capable of 1440p gaming with all the bells and whistles. We're talking RTX, NVIDIA hairworks, faceworks and whatever else works. The latter two are not NVIDIA features, please don't write news pieces about them.

SPECIFICATIONS

The RTX 4070 is built on NVIDIA's Ada Lovelace architecture. Ada Lovelace introduces several new features and improvements, such as better ray tracing capabilities, faster rendering, and improved power efficiency. The GPU configuration of the RTX 4070 is divided into four Graphics Processing Clusters (GPCs). Each GPC serves as a high-level organisational unit which includes Streaming Multiprocessors (SMs), Texture Processing Clusters (TPCs), and Raster Operating Units (ROPs). In total, there are 46 SMs on the GPU, which are responsible for executing shader programs and handling various graphics processing tasks. Each SM features 128 CUDA Cores. In addition to the CUDA Cores, the RTX 4070 is equipped with 184 Tensor Cores, 4 per SM, which are specialised hardware units designed to accelerate AI-based computations. The RTX 4070 also includes 46 RT Cores, one per SM, which are dedicated ray tracing hardware units. With these dedicated RT Cores, the GPU can perform ray tracing tasks much faster. The GPU has 184 Texture Units, which are responsible for handling texture mapping tasks. Furthermore, the RTX 4070 features 64 ROP Units, which handle the final



stage of rendering, including anti-aliasing and outputting pixel data to the frame buffer. The card comes with a sizable 36 MB of L2 cache. The RTX 4070 is equipped with 12 GB of GDDR6X memory, which operates on a 192-bit memory bus. The narrower bus width is fine since the L2 cache is large. Lastly, the RTX 4070 supports AV1 hardware encoding, a next-generation video codec. This hardware support ensures that the GPU can efficiently encode and decode AV1 video streams, providing better playback and streaming capabilities. OBS has already incorporated support for AV1 codecs.

BUILD QUALITY

The main PCB is quite compact and takes up very little volume and there's a massive metal heatsink fin array spanning the entirety of the body. The 4070 has a twin fan assembly which is configured to optimise maximum airflow through the fin assembly. There's little to say about the build quality. What is interesting, though is that the RTX 4070 is a much smaller card than the other RTX 40 series cards and almost feels like a 60-class card.

PERFORMANCE

The RTX 4070 is just a smidge below the RTX 3090 in 3DMark Time Spy and a little ahead of the RTX 3080. As for the competition, the AMD RX 6950 XT is a

teensy bit ahead and the newly launched 7900 XTX is miles ahead. As for the performance of the various APIs, the RTX 4070 is ahead of the 6950 XT in OpenGL titles because AMD cards of the RDNA 2 generation have been at a disadvantage.

However, the RDNA 3 cards no longer have the handicap and the 7900 XTX outperforms the 4070 across all APIs. Compared to its own previous gen RTX 3080 and 3090, the 4070 lags a little behind. In video games, it goes head to head with the RTX 3080 in a lot of video games. Even with RTX switched on, the two cards are evenly matched. That being said, the 40-series cards have support for DLSS 3 and that's an advantage which will see the performance gap widen over time. Lastly, the noise and temperature levels are very much in check with peak temperatures barely crossing into the mid 70s while gaming. Power levels, on the other hand, are much lesser than the RTX 30 series.

VERDICT

The NVIDIA GeForce RTX 4070 is a pretty good upgrade for folks coming in from an RTX 20-series card or older. Currently, the RTX 3080 is retailing for INR 66,000 in India so the RTX 4070 easily surpasses the RTX 3080 considering that it is equal in performance in most games without RTX or even when running DLSS 2 along with RTX. The key differentiators are the improved RT Cores, Tensor Cores and improved power efficiency thanks to the TSMC 4N process that it is build on. You can easily say that the RTX 3080 has been effectively replaced by the RTX 4070.

—Mithun Mohandas

SPECIFICATIONS

Boost Clock (GHz): 2.48 | Max Display Resolution: 4K at 240Hz or 8K at 60Hz with DSC | Ray Tracing Cores: 3rd Generation | Tensor Cores: 4th Generation | NVIDIA Architecture: Ada Lovelace

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